Week-3 **Seller and Inventory Management**

Scenario:

The Seller and Inventory Management System is designed for an e-commerce platform that manages product supplies from multiple independent sellers/vendors. The system records detailed vendor profile information, tracks products supplied by each vendor, and dynamically manages stock levels to prevent stockouts and optimize fulfillment.

1. **Create Seller and Inventory Tables:**

```
SQL> -- Create Seller Table
SQL> CREATE TABLE Seller_Details (
  2      Seller_Code VARCHAR2(8),
  3      Vendor_Name VARCHAR2(30),
  4      Contact_Email VARCHAR2(35),
  5      Phone_No NUMBER(10),
  6      Store_Address VARCHAR2(50)
  7  );

Table created.

SQL>
SQL> -- Create Inventory Table
SQL> CREATE TABLE Inventory_Control (
  2      Inv_Ref_ID VARCHAR2(8),
  3      Item_ID VARCHAR2(8),
  4      Seller_Code VARCHAR2(8),
  5      Qty_In_Hand NUMBER,
  6      Status VARCHAR2(12),
  7      Last_Updated_On DATE
  8  );

Table created.
```

2. Establish Relationships

Seller Details and Product

- **Relationship:** One seller can supply multiple products. One product can be supplied by one or multiple sellers.
- **Mapping**: Seller_Details → Product is one to many**.**

Product and Inventory Control

- **Relationship:** Each product has inventory information. Inventory_Control tracks available stock**.**

- **Mapping:** Product → Inventory_Control is one to one.

Primary Key added for Both Tables:

```
SQL> ALTER TABLE Seller_Details
  2  ADD CONSTRAINT PK_SellerDetails PRIMARY KEY (Seller_Code);

Table altered.

SQL>
SQL> ALTER TABLE Inventory_Control
  2  ADD CONSTRAINT PK_InvControl PRIMARY KEY (Inv_Ref_ID);

Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory_Control
  2  ADD CONSTRAINT FK_Inv_Seller
  3  FOREIGN KEY (Seller_Code) REFERENCES Seller_Details (Seller_Code);

Table altered.

SQL>
SQL> ALTER TABLE Inventory_Control
  2  ADD CONSTRAINT FK_Inv_Product
  3  FOREIGN KEY (Item_ID) REFERENCES Product (Product_ID);

Table altered.
```

Unique and Not Null Constraints:

```
SQL> ALTER TABLE Seller_Details
  2  ADD CONSTRAINT UNQ_Seller_Email UNIQUE (Contact_Email);

Table altered.

SQL>
SQL> ALTER TABLE Seller_Details
  2  MODIFY Contact_Email NOT NULL;

Table altered.
```

Check Constraint:

```
SQL> ALTER TABLE Inventory_Control
  2  ADD CONSTRAINT CK_Qty_NonNegative CHECK (Qty_In_Hand >= 0);

Table altered.
```

3. Insert Values into Tables

Insert into Seller_Details Table

```
SQL> INSERT INTO Seller VALUES ('S00001', 'GlowCare Suppliers', 'glowcare@gmail.com', '9876543210', '12, Anna Nagar, Chennai');

1 row created.

SQL> INSERT INTO Seller VALUES ('S00002', 'Luxe Beauty Corp', 'contact@luxebeauty.com', '9876543012', '45, MG Road, Bengaluru');

1 row created.

SQL> INSERT INTO Seller VALUES ('S00003', 'Aura Botanicals', 'aurabotanicals@gmail.com', '7890654312', '88, T. Nagar, Chennai');

1 row created.

SQL> INSERT INTO Seller VALUES ('S00004', 'Velvet Glam Distro', 'velvetglam@gmail.com', '7824003009', '23, Cross Street, Mumbai');

1 row created.
```

```
SQL> SELECT * FROM Seller;

SELLER_ SELLER_NAME
------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
S00001  GlowCare Suppliers
glowcare@gmail.com                                 9876543210
12, Anna Nagar, Chennai

S00002  Luxe Beauty Corp
contact@luxebeauty.com                             9876543012
45, MG Road, Bengaluru

SELLER_ SELLER_NAME
------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------

S00003  Aura Botanicals
aurabotanicals@gmail.com                           7890654312
88, T. Nagar, Chennai

S00004  Velvet Glam Distro
velvetglam@gmail.com                               7824003009

SELLER_ SELLER_NAME
------- --------------------------------------------------
EMAIL                                              PHONE
-------------------------------------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
23, Cross Street, Mumbai
```

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES ('Inv0001', 'PRD001', 'S00001', 25, 'Available', TO_DATE('2026-09-12', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES ('Inv0002', 'PRD001', 'S00002', 15, 'Available', TO_DATE('2026-08-13', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES ('Inv0003', 'PRD002', 'S00003', 0, 'Unavailable', TO_DATE('2026-09-15', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO Inventory VALUES ('Inv0004', 'PRD004', 'S00004', 30, 'Available', TO_DATE('2026-09-14', 'YYYY-MM-DD'));

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INV_ID  PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
------- ------- ------- -------------- --------------- ---------
Inv0001 PRD001  S00001              25 Available       12-SEP-26
Inv0002 PRD001  S00002              15 Available       13-AUG-26
Inv0003 PRD002  S00003               0 Unavailable     15-SEP-26
Inv0004 PRD004  S00004              30 Available       14-SEP-26
```

3. Execution of System Reports

Report 1: Seller-wise Product Report

Objective

This report aggregates inventory data to evaluate each seller's contribution by displaying their total product varieties supplied and total current stock quantities in the warehouse.

Required Display Fields

1. Seller Name: The full registered name of the vendor/seller.
2. Number of Products Supplied: The total count of distinct products linked to the seller.
3. Available Stock: The total sum of units available across all products for that seller.

```
SQL> SELECT s.Seller_Name,
  2         COUNT(i.Product_ID) AS Number_of_Products,
  3         SUM(i.Stock_Quantity) AS Available_Stock
  4  FROM Seller s
  5  JOIN Inventory i ON s.Seller_ID = i.Seller_ID
  6  WHERE i.Stock_Status = 'Available'
  7  GROUP BY s.Seller_Name;

SELLER_NAME                                        NUMBER_OF_PRODUCTS
-------------------------------------------------- ------------------
AVAILABLE_STOCK
---------------
Velvet Glam Distro                                                  1
             30

Luxe Beauty Corp                                                    1
             15

GlowCare Suppliers                                                  1
             25
```

4.Use Joins to get details from Seller and Inventory Table.